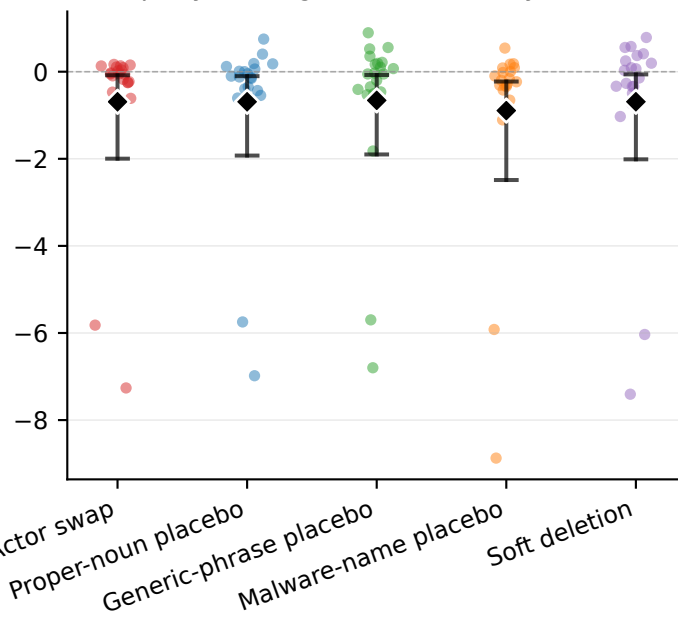


Step D primary inferential test: paired Δ -margin DiD

Per-query mean Δ -margin (perturbed – original)

(a) Per-query Δ -margin distribution by intervention



(b) Paired DiD: swap vs placebo (BCa 95% CI)

